Mario Alan Soto Delgado

Aerospace Structural Engineer

Stress & MRB Engineering | Aircraft Repair & Substantiation | Flight Dynamics

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Education

Autonomous University of Chihuahua, School of Engineering — **Chihuahua, México**

B.S. Aerospace Engineering — 2013 - 2017

B.S. Engineering Physics — 2015 - 2019

Experience

Altitude Aerospace USA — **Portland, Oregon / Dallas, Texas / Remote**

Stress Liaison Engineer — **Nov 2023 - Jun 2026**

- Worked through the certification process of the first passenger-to-freighter conversion of a Boeing 777 by providing stress support to over 300 aircraft deviations.
- Trained 3 junior engineers into the fundamentals of structural analysis and deviations assessment.
- Audited, wrote and updated certification reports that covered: main cargo door and surrounding assemblies, floor, pressure deck, overwing box, shear panels and systems reinforcements. All of them obtained FAA approval.
- Developed over 20 tools for quick assessment of aircraft deviations on Airbus A321 undergoing a passenger to cargo conversion.

Textron Aviation Mexico (American Industries) — **Chihuahua, México**

MRB Engineer — **Jun 2022 - Oct 2023**

- Conducted static structural analysis to assess integrity and repairability of over 800 non-conforming aircraft components for the SkyCourier, Longitude and XLS Cessna aircrafts.
- Collaborated with fatigue and damage tolerance, and materials and processes teams on high-risk cases.
- Developed and maintained weekly KPI metrics to support operational decision-making.

Chihuahua Institute of Technology — **Chihuahua, México**

Teacher in the Fundamental Sciences Department — **Mar 2022 - Jul 2022**

- Taught classes for students in the following engineering programs: mechanical, electromechanical, material science and industrial/manufacturing. Courses: Thermodynamics and Integral Calculus.

Scale AI (Remotasks) — **Remote**

Quality Assurance Specialist — **Oct 2021 - Jun 2022**

- Performed quality audits on web scraping processes for e-commerce platforms.
- Led root-cause analysis initiatives to identify and solve inconsistencies with quality standards.
- Collaborated cross-functionally to implement corrective actions and improve data reliability.

Autonomous University of Chihuahua, School of Engineering — **Chihuahua, México**

Administrative Assistant at the Engineering Physics & Mathematics Department — **Jan 2018 - Dec 2019**

- Performed basic statistical analysis and managed the evidence archive for the certification process of nine educational programs. All of them got certified.
- Addressed emerging requests by providing solutions to academic and administrative problems.

Projects

Mexican Nanosatellite for Renewable Energies Development (NSATMX-ER1) — **Remote**

Flight Dynamics Mission Analyst — **Sept 2020 - May 2021**

- Conducted coverage and age-of-data analysis for a hyperspectral imaging Earth observation satellite using MATLAB and STK.
- Produced technical reports and data visualizations of results to support mission design decisions.
- Aided in the definition of objectives, constraints, and requirements (OCR) for the communications subsystem.

AzTechSat-1 (UPAEP / Mexico's Space Agency / NASA) **Puebla, México / Remote**
Technical Team Leader of the Attitude Determination and Control Subsystem (ADCS) **Jul 2017 - Aug 2019**

- Led a multidisciplinary engineering team in preparation for NASA readiness reviews.
- Directed over 1,200 simulations of an ADCS model in MATLAB to determine optimal flight configuration. The model included a J2 orbit propagator, the IGRF geo-magnetic model, a b-dot controller and attitude representation by quaternions.
- Coordinated with hardware providers to validate integration procedures.
- Presented technical results to NASA evaluation committees.

Other Activities

NASA's International Space Apps Challenge **2019 - Present (Chihuahua, Mexico / Remote)**
Chihuahua's Local Lead *Jun 2019 - Dic 2023*

- Managed logistics, budget, and resource allocation for an event that coordinated the efforts of over 200 people each year, considering the organizing team, participants, mentors, judges and sponsors.
- Led public outreach and community engagement initiatives.

NASA Space Apps Collective – 1st Generation Cohort Member *Sep 2023 - Jun 2024*

- Participated in a series of workshops regarding Diversity, Equity, Inclusion and Accessibility initiatives.
- Participated in the Collective Genius Summit in March 2024 with the Lightning Talk: "Inspiring Communities".

Navigator & Global Judge *Jun 2025 - Present*

- Helped teams across the globe to improve their project submission by aligning them to the evaluation criteria.
- Audited challenge resources and evaluated project submissions to determine the global finalists selection.

Space Generation Advisory Council **2020 - 2025**
Delegate in the Working Group: "Education and Professional Development in the Space Sector". *Dec 2020*

- Participated in a series of online discussion groups to identify the technological and educational capabilities of students and young professionals in the North America, Central America and the Caribbean (NCAC) region.

Delegate in the Working Group: "Space Mission Design for Sustainable Development Goals". *Sept 2023*

- Along with an interdisciplinary team, we developed a proposal to use artificial intelligence and satellite imaging to identify optimal water distribution routes for remote communities across the NCAC region.
- Workshop held in San José, Costa Rica.

Delegate in the Working Group: "Space Entrepreneurship". *Jul 2024*

- Participated in a series of discussion groups with space entrepeuners of the NCAC to identify the biggest challenges of developing commercial space technology in the region.
- Workshop held in Puebla, México.

Organizing Team Member, 5th North and Central American Space Generation Workshop *May 2025*

- Worked along an international organizing team and representatives from Canada's Space Agency to develop a series of workshops and panels to further develop collaboration among space professionals in the region.
- Responsible for the logistics of the event: volunteers coordination, participants registration and time management of the event's agenda.
- Workshop held in Montreal, Canada.

Autonomous University of Chihuahua, School of Engineering **2018 (Chihuahua, Mexico)**
Course Instructor: "Introduction to MATLAB" *Aug 2018 - Oct 2018*

- Prepared and taught an introductory course to the basic programming tools in MATLAB.
- Topics covered: matrices and systems of equations, cycles and conditionals, two-dimensional plots, surface plots, function programming and numerical solution of differential equations.

Awards & Achievements

Sept 2022 **GalileoGICkathon 2022 (Galileo Information Center) - 3rd Place Winner**
Project proposal for an AI-based security system for tracking and preventing criminal activities affecting the transportation/logistics industry using Galileo GNSS data.

Aug 2020 **ASCLEPIOS Hackathon (École Polytechnique Fédérale de Lausanne)**
Development of a proposal for a lunar mining system using geo-physical methods.

May 2020 **Mexico's Space Missions 2020: COVID-19 Challenge (Mexico's Space Agency) - 2nd Place Winner**
Winner of the second place in the category of *support to the most vulnerable groups* with the project: "Correlation of gender violence in vulnerable sectors in Mexico during lockdown due to COVID-19".

Oct 2017 **LX National Congress of Physics (Mexican Society of Physics)**
Presentation of the project: "Simulation of a Positioning in Orbit and Attitude Analysis of a 1U CubeSat".

Courses & Certifications

Jul 2020 **Certified SolidWorks Associate (CSWA)**
Dassault Systèmes [ID: C-ZTAM3JESNF]

Feb 2021 **Hyperspectral Data for Land and Coastal Systems**
NASA's Applied Remote Sensing Training Program

Skills

Languages Spanish (Native), English (Professional Proficiency).

Engineering Structural Analysis, Aircraft Repair Evaluation, Simulation, Orbital Mechanics.

Tools MATLAB, Siemens NX, SolidWorks, Excel, STK.

Core Strengths Technical Leadership, Problem-solving, Cross-functional Coordination, Independent Learning.